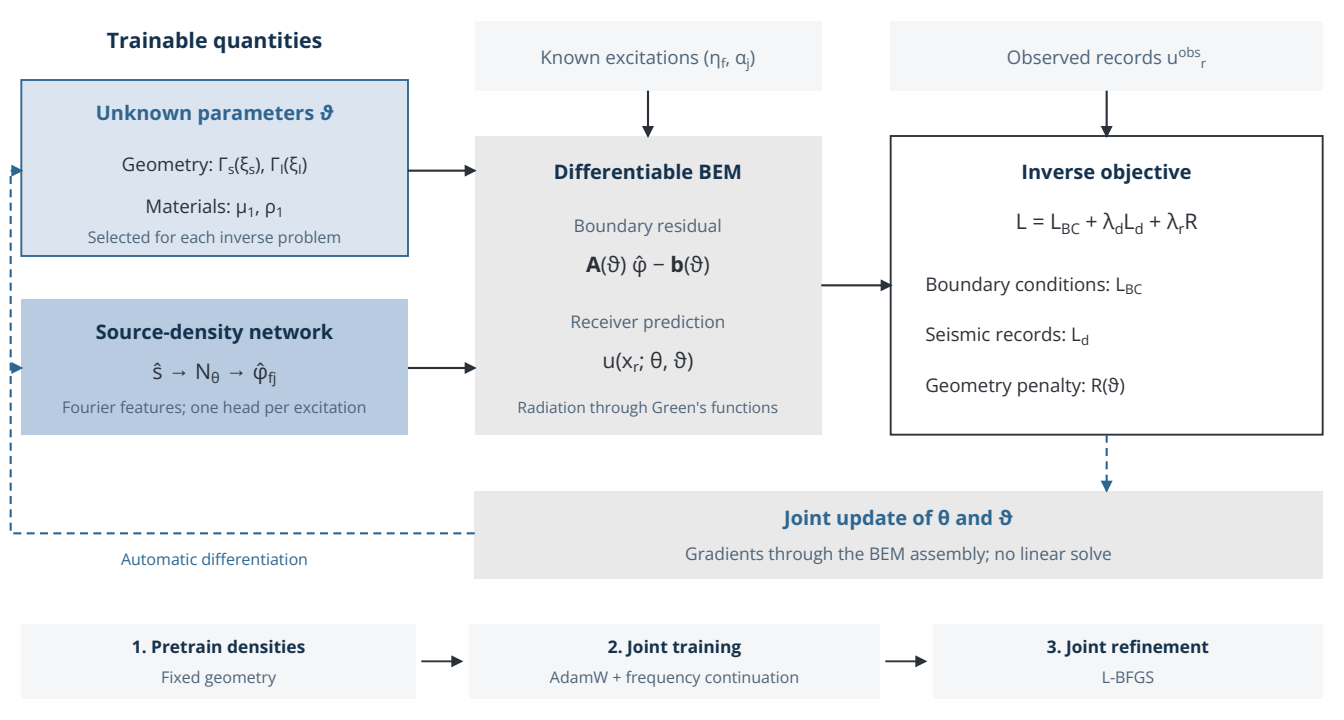
Trainable quantities
Unknown parameters ϑ
Geometry: Γs(ξs), Γl(ξl)
Materials: μ1, ρ1
Selected for each inverse problem
Source-density network
ŝ → Nθ → φ̂fj
Fourier features; one head per excitation
Known excitations (ηf, αj)
Differentiable BEM
Boundary residual
A(ϑ) φ̂ − b(ϑ)
Receiver prediction
u(xr; θ, ϑ)
Radiation through Green's functions
Observed records u^obs_r
Inverse objective
L = LBC + λdLd + λrR
Boundary conditions: LBC
Seismic records: Ld
Geometry penalty: R(ϑ)
Joint update of θ and ϑ
Gradients through the BEM assembly; no linear solve
Automatic differentiation
1. Pretrain densities
Fixed geometry
2. Joint training
AdamW + frequency continuation
3. Joint refinement
L-BFGS